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ADAPTIVE DISPLAY OF AN ON-SCREEN RETICLE

Abstract

A display device creates an off-game hardware reticle and obtains red, green, and blue (RGB) values of pixels, sent by an information handling system from an area underneath the reticle. If the difference between an average of the RGB values and an RGB value of the reticle is less than a threshold, then the display device changes the RGB value of the reticle to a complementary color.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure generally relates to information handling systems, and more particularly relates to an adaptive display of an on-screen reticle.

BACKGROUND

[0002] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs and requirements can vary between different applications. Thus, information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling systems can also implement various virtualized architectures. Data and voice communications among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

[0003] A display device creates an off-game hardware reticle and obtains red, green, and blue (RGB) values of pixels, sent by an information handling system from an area underneath the reticle. If the difference between an average of the RGB values and an RGB value of the reticle is less than a threshold, then the display device changes the RGB value of the reticle to a complementary color.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:

[0005] FIG. 1 is a block diagram illustrating an information handling system according to an embodiment of the present disclosure;

[0006] FIG. 2 is a block diagram of a system configured for adaptively displaying an on-screen reticle, according to an embodiment of the present disclosure;

[0007] FIG. 3 is a flowchart illustrating a method for adaptive display of an on-screen reticle, according to an embodiment of the present disclosure;

[0008] FIGS. 4 and 5 are block diagrams of reticles superimposed over images, according to an embodiment of the present disclosure;

[0009] FIG. 6 is a block diagram of a system configured for adaptive display of an on-screen reticle, according to an embodiment of the present disclosure;

[0010] FIGS. 7 and 8 are block diagrams of composites that include a series of images for an adaptive display of an on-screen reticle, according to an embodiment of the present disclosure;

[0011] FIG. 9 is a flowchart illustrating a method for adaptive display of an on-screen reticle, according to an embodiment of the present disclosure; and

[0012] FIGS. 10 and 11 are block diagrams of reticles superimposed over images, according to an embodiment of the present disclosure.

[0013] The use of the same reference symbols in different drawings indicates similar or identical

items.

DETAILED DESCRIPTION OF THE DRAWINGS

[0014] The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

[0015] FIG. 1 illustrates an embodiment of an information handling system **100** including processors **102** and **104**, a chipset **110**, a memory **120**, a graphics adapter **130** connected to a video display **134**, a non-volatile RAM (NV-RAM) **140** that includes a basic input and output system/extensible firmware interface (BIOS/EFI) module **142**, a disk controller **150**, a hard disk drive (HDD) **154**, an optical disk drive **156**, a disk emulator **160** connected to a solid-state drive (SSD) **164**, an input/output (I/O) interface **170** connected to an add-on resource **174** and a trusted platform module (TPM) **176**, a network interface **180**, and a baseboard management controller (BMC) **190**. Processor **102** is connected to chipset **110** via processor interface **106**, and processor **104** is connected to the chipset via processor interface **108**. In a particular embodiment, processors **102** and **104** are connected together via a high-capacity coherent fabric, such as a HyperTransport link, a QuickPath Interconnect, or the like. Chipset **110** represents an integrated circuit or group of integrated circuits that manage the data flow between processors **102** and **104** and the other elements of information handling system **100**. In a particular embodiment, chipset **110** represents a pair of integrated circuits, such as a northbridge component and a southbridge component. In another embodiment, some or all of the functions and features of chipset **110** are integrated with one or more of processors **102** and **104**.

[0016] Memory **120** is connected to chipset **110** via a memory interface **122**. An example of memory interface **122** includes a Double Data Rate (DDR) memory channel and memory **120** represents one or more DDR Dual In-Line Memory Modules (DIMMs). In a particular embodiment, memory interface **122** represents two or more DDR channels. In another embodiment, one or more of processors **102** and **104** include a memory interface that provides a dedicated memory for the processors. A DDR channel and the connected DDR DIMMs can be in accordance with a particular DDR standard, such as a DDR3 standard, a DDR4 standard, a DDR5 standard, or the like.

[0017] Memory **120** may further represent various combinations of memory types, such as Dynamic Random Access Memory (DRAM) DIMMs, Static Random Access Memory (SRAM) DIMMs, non-volatile DIMMs (NV-DIMMs), storage class memory devices, Read-Only Memory (ROM) devices, or the like. Graphics adapter **130** is connected to chipset **110** via a graphics interface **132** and provides a video display output **136** to a video display **134**. An example of a graphics interface **132** includes a Peripheral Component Interconnect-Express (PCIe) interface and graphics adapter **130** can include a four-lane ($\times 4$) PCIe adapter, an eight-lane ($\times 8$) PCIe adapter, a 16-lane ($\times 16$) PCIe adapter, or another configuration, as needed or desired. In a particular embodiment, graphics adapter **130** is provided down on a system printed circuit board (PCB). Video display output **136** can include a Digital Video Interface (DVI), a High-Definition Multimedia Interface (HDMI), a DisplayPort interface, or the like, and video display **134** can include a monitor, a smart television, an embedded display such as a laptop computer display, or the like.

[0018] NV-RAM **140**, disk controller **150**, and I/O interface **170** are connected to chipset **110** via an I/O channel **112**. An example of I/O channel **112** includes one or more point-to-point PCIe links between chipset **110** and each of NV-RAM **140**, disk controller **150**, and I/O interface **170**. Chipset **110** can also include one or more other I/O interfaces, including a PCIe interface, an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I.sup.2C) interface, a System Packet Interface (SPI), a Universal Serial Bus

(USB), another interface, or a combination thereof. NV-RAM **140** includes BIOS/EFI module **142** that stores machine-executable code (BIOS/EFI code) that operates to detect the resources of information handling system **100**, to provide drivers for the resources, to initialize the resources, and to provide common access mechanisms for the resources. The functions and features of BIOS/EFI module **142** will be further described below.

[0019] Disk controller **150** includes a disk interface **152** that connects the disc controller to a hard disk drive (HDD) **154**, to an optical disk drive (ODD) **156**, and to disk emulator **160**. An example of disk interface **152** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **160** permits SSD **164** to be connected to information handling system **100** via an external interface **162**. An example of external interface **162** includes a USB interface, an institute of electrical and electronics engineers (IEEE) **1394** (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, SSD **164** can be disposed within information handling system **100**.

[0020] I/O interface **170** includes a peripheral interface **172** that connects the I/O interface to add-on resource **174**, to TPM **176**, and to network interface **180**. Peripheral interface **172** can be the same type of interface as I/O channel **112** or can be a different type of interface. As such, I/O interface **170** extends the capacity of I/O channel **112** when peripheral interface **172** and the I/O channel are of the same type, and the I/O interface translates information from a format suitable to the I/O channel to a format suitable to the peripheral interface **172** when they are of a different type. Add-on resource **174** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource **174** can be on a main circuit board, on separate circuit board, or add-in card disposed within information handling system **100**, a device that is external to the information handling system, or a combination thereof.

[0021] Network interface **180** represents a network communication device disposed within information handling system **100**, on a main circuit board of the information handling system, integrated onto another component such as chipset **110**, in another suitable location, or a combination thereof. Network interface **180** includes a network channel **182** that provides an interface to devices that are external to information handling system **100**. In a particular embodiment, network channel **182** is of a different type than peripheral interface **172** and network interface **180** translates information from a format suitable to the peripheral channel to a format suitable to external devices.

[0022] In a particular embodiment, network interface **180** includes a NIC or host bus adapter (HBA), and an example of network channel **182** includes an InfiniBand channel, a Fibre Channel, a Gigabit Ethernet channel, a proprietary channel architecture, or a combination thereof. In another embodiment, network interface **180** includes a wireless communication interface, and network channel **182** includes a Wi-Fi channel, a near-field communication (NFC) channel, a Bluetooth® or Bluetooth-Low-Energy (BLE) channel, a cellular based interface such as a Global System for Mobile (GSM) interface, a Code-Division Multiple Access (CDMA) interface, a Universal Mobile Telecommunications System (UMTS) interface, a Long-Term Evolution (LTE) interface, or another cellular based interface, or a combination thereof. Network channel **182** can be connected to an external network resource (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.

[0023] BMC **190** is connected to multiple elements of information handling system **100** via one or more management interface **192** to provide out of band monitoring, maintenance, and control of the elements of the information handling system. As such, BMC **190** represents a processing device different from processor **102** and processor **104**, which provides various management functions for

information handling system **100**. For example, BMC **190** may be responsible for power management, cooling management, and the like. The term BMC is often used in the context of server systems, while in a consumer-level device, a BMC may be referred to as an embedded controller (EC). A BMC included in a data storage system can be referred to as a storage enclosure processor. A BMC included at a chassis of a blade server can be referred to as a chassis management controller and embedded controllers included at the blades of the blade server can be referred to as blade management controllers. Capabilities and functions provided by BMC **190** can vary considerably based on the type of information handling system. BMC **190** can operate in accordance with an Intelligent Platform Management Interface (IPMI). Examples of BMC **190** include an Integrated Dell® Remote Access Controller (iDRAC).

[0024] Management interface **192** represents one or more out-of-band communication interfaces between BMC **190** and the elements of information handling system **100**, and can include an Inter-Integrated Circuit (I2C) bus, a System Management Bus (SMBUS), a Power Management Bus (PMBUS), a Low Pin Count (LPC) interface, a serial bus such as a Universal Serial Bus (USB) or a Serial Peripheral Interface (SPI), a network interface such as an Ethernet interface, a high-speed serial data link such as a PCIe interface, a Network Controller Sideband Interface (NC-SI), or the like. As used herein, out-of-band access refers to operations performed apart from a BIOS/operating system execution environment on information handling system **100**, that is apart from the execution of code by processors **102** and **104** and procedures that are implemented on the information handling system in response to the executed code.

[0025] BMC **190** operates to monitor and maintain system firmware, such as code stored in BIOS/EFI module **142**, option ROMs for graphics adapter **130**, disk controller **150**, add-on resource **174**, network interface **180**, or other elements of information handling system **100**, as needed or desired. In particular, BMC **190** includes a network interface **194** that can be connected to a remote management system to receive firmware updates, as needed or desired. Here, BMC **190** receives the firmware updates, stores the updates to a data storage device associated with the BMC, transfers the firmware updates to NV-RAM of the device or system that is the subject of the firmware update, thereby replacing the currently operating firmware associated with the device or system, and reboots information handling system, whereupon the device or system utilizes the updated firmware image.

[0026] BMC **190** utilizes various protocols and application programming interfaces (APIs) to direct and control the processes for monitoring and maintaining the system firmware. An example of a protocol or API for monitoring and maintaining the system firmware includes a graphical user interface (GUI) associated with BMC **190**, an interface defined by the Distributed Management Taskforce (DMTF) (such as a Web Services Management (WSMan) interface, a Management Component Transport Protocol (MCTP) or, a Redfish® interface), various vendor defined interfaces (such as a Dell EMC Remote Access Controller Administrator (RACADM) utility, a Dell EMC OpenManage Enterprise, a Dell EMC OpenManage Server Administrator (OMSS) utility, a Dell EMC OpenManage Storage Services (OMSS) utility, or a Dell EMC OpenManage Deployment Toolkit (DTK) suite), a BIOS setup utility such as invoked by a “F2” boot option, or another protocol or API, as needed or desired.

[0027] In a particular embodiment, BMC **190** is included on a main circuit board (such as a baseboard, a motherboard, or any combination thereof) of information handling system **100** or is integrated onto another element of the information handling system such as chipset **110**, or another suitable element, as needed or desired. As such, BMC **190** can be part of an integrated circuit or a chipset within information handling system **100**. An example of BMC **190** includes an iDRAC, or the like. BMC **190** may operate on a separate power plane from other resources in information handling system **100**. Thus BMC **190** can communicate with the management system via network interface **194** while the resources of information handling system **100** are powered off. Here, information can be sent from the management system to BMC **190** and the information can be

stored in a RAM or NV-RAM associated with the BMC. Information stored in the RAM may be lost after power-down of the power plane for BMC **190**, while information stored in the NV-RAM may be saved through a power-down/power-up cycle of the power plane for the BMC.

[0028] Information handling system **100** can include additional components and additional busses, not shown for clarity. For example, information handling system **100** can include multiple processor cores, audio devices, and the like. While a particular arrangement of bus technologies and interconnections is illustrated for the purpose of example, one of skill will appreciate that the techniques disclosed herein are applicable to other system architectures. Information handling system **100** can include multiple central processing units (CPUs) and redundant bus controllers. One or more components can be integrated together. Information handling system **100** can include additional buses and bus protocols, for example, I2C and the like. Additional components of information handling system **100** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display.

[0029] For purposes of this disclosure information handling system **100** can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **100** can be a personal computer, a laptop computer, a smartphone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch, a router, or another network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **100** can include processing resources for executing machine-executable code, such as processor **102**, a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **100** can also include one or more computer-readable media for storing machine-executable code, such as software or data.

[0030] Off-game hardware reticles, such as crosshairs, are created to assist game players to aim and target game objects. For some implementations, in-game crosshairs have a single static color for the duration of a game. Because of this, the in-game crosshair may become difficult to distinguish from the backdrop of a scene as the color of the backdrop approaches the color of the in-game crosshair. Other game implementations include a backdrop that can overlap with the in-game crosshairs, thus hinder inga user's view of the gameplay. To address these and other concerns, the present disclosure provides a system and method for a real-time on-screen enhancement that maintains a consistently visible off-game hardware reticle both during the gameplay and outside of games without adding to CPU or graphics processing unit (GPU) rendering time. In particular, instead of the crosshair having a static single color, the present disclosure provides a system and method that is configured to provide an adaptive color-changing crosshair.

[0031] FIG. 2 shows a system **200** configured for adaptive display of an on-screen reticle. System **200** includes an information handling system **205** that is coupled to a display device **240** via an interface **290**. Information handling system **205**, which is similar to information handling system **100** of FIG. 1, includes a memory **210**, a CPU **220**, and a GPU **225**. Memory **210** includes an operating system **215**. Display device **240** includes a display panel **245** and an interface board **260**. Display panel **245** includes a timing control unit **250** while interface board **260** includes a scalar **265**. The components of system **200** may be implemented in hardware, software, firmware, or any combination thereof. The components shown are not drawn to scale and system **200** may include additional or fewer components. In addition, connections between components may be omitted for descriptive clarity.

[0032] Information handling system **205** may be configured to generate visual information for display at display device **240**. For example, CPU **220** may be configured to execute instructions

from operating system **215** or another application. CPU **220** may also be configured to process information stored in memory **210**. Memory **210**, which is similar to memory **120** of FIG. **1**, maybe a non-volatile memory that is configured to store operating system **215**. Operating system **215** may be loaded during a boot process and enter a runtime environment. CPU **220**, which is similar to processors **102** and **104**, may be configured to execute code retrieved from memory **210**, such as operating system **215**. Although information handling system **205** is illustrated with a single CPU, other embodiments may each be configured identically, or to provide specialized processing operations.

[0033] CPU **220** may include any processor capable of executing program instructions. CPU **220** may be communicatively coupled to operating system **215** and GPU **225**. Accordingly, GPU **225** may be configured to interface with CPU **220** to process information for communication to display device **240**. Generally, GPU **225** generates pixel values that pixels at display panel **245** present as colors so that the accumulation of the pixels presents a visual image. GPU **225** is coupled to display device **240** via interface **290**. Interface **290** can be an HDMI, DisplayPort™, USB, USB type-C, video graphics array (VGA), or similar interface. However, any variety of connections between GPU **225** and scalar **265** are envisioned as falling within the scope of the present disclosure.

[0034] Display device **240** may be integrated into information handling system **205**, such as where information handling system **205** is a laptop, tablet, 2-in-1 convertible device, mobile device, or similar. Display device **240** may also be an external display device that is coupled to information handling system **205**, such as a computer monitor. Display device **240** includes an interface board **260** that is configured to receive image data from GPU **225**, process the received data via scalar **265**, and transmit the processed image data to display panel **245**. Display panel **245** may be a light-emitting diode (LED) panel, an organic LED panel, a liquid crystal display panel, or other display technologies. Display panel **245** includes a timing control unit **250** that may be configured to receive and process pixel data from scalar **265**.

[0035] Interface board **260** includes a scalar **265** that may be configured to convert different video signals, such as HDMI, DisplayPort™, USB-C, and VGA video signals, into a format that can be used to render content on display panel **245**. Scalar **265** may include image processing capabilities to manipulate how the content is rendered. Scalar **265** includes a dynamic contrast ratio (DCR) engine **270**, a microcontroller unit (MCU) engine **280**, and an on-screen display (OSD) engine **275**. DCR engine **270** may be configured to utilize local zone chromaticity analysis to analyze content, such as an image or frame. In particular, DCR engine **270** may be configured to collect red, green, and blue (RGB) values from pixels beneath a reticle, such as a crosshair, and calculate one or more average values from the collected RGB values, which provide an understanding of the visual context beneath the reticle.

[0036] Subsequently, MCU engine **280** may be configured to process the average value(s) of the RGB values and conduct a comparison against predefined conditions to make informed decisions regarding crosshair color adaptation. For example, MCU engine **280** may compare the average RGB value of an image or video frame with the RGB value of the reticle, and MCU engine **280** may determine if it is desirable to change the RGB value of the reticle or not. If MCU engine **280** determines that it is desirable to change the color of the reticle, MCU engine **280** may trigger OSD engine **275** to perform an action. Otherwise, MCU engine **280** may perform another comparison. OSD engine **275** may be configured to update the color of the reticle based on the visual context underneath when triggered. In particular, OSD engine **275** may be configured to adaptively transform the color of the reticle into a complementary color of the average RGB value over the image or video frame. This allows the user to see the crosshair against the backdrop in real-time. As such, as the crosshair moves during gameplay, the color of the crosshair may change adaptively based on the colors of a scene on display.

[0037] Those of ordinary skill in the art will appreciate that the configuration, hardware, and/or

software components of system **200** depicted in FIG. **2** may vary. For example, the illustrative components within system **200** are not intended to be exhaustive but rather are representative to highlight components that can be utilized to implement aspects of the present disclosure. For example, other devices and/or components may be used in addition to or in place of the devices/components depicted. The depicted example does not convey or imply any architectural or other limitations with respect to the presently described embodiments and/or the general disclosure. In the discussion of the figures, reference may also be made to components illustrated in other figures for continuity of the description.

[0038] FIG. **3** shows a flowchart of a method **300** for an adaptive display of an on-screen reticle. Method **300** may be performed by any suitable component of system **200** of FIG. **2**, including, but not limited to, GPU **225** and scalar **265**. In particular, GPU **225** may be configured to perform block **315** while scalar **265** may be configured to perform the rest. While embodiments of the present disclosure are described in terms of the components of system **200** of FIG. **2**, it should be recognized that other components may be utilized to perform the described method. Method **300** may be performed while an adaptive reticle function over a video frame is enabled by a user.

[0039] Method **300** typically starts at block **310** where the user may enable or activate the reticle function over the video frame. This enables the reticle to be overlaid on the content, such as an image or video frame, and succeeding contents until deactivated by the user. At this point, the user may choose or adjust one or more settings associated with the reticle, such as the location of the reticle, which by default is the center. Other settings include an initial color and shape of the reticle. In one example, the default color of the reticle is green with an RGB (0, 255, 0) while the default shape is a crosshair with a bar-cross shape which is represented as a pair of perpendicularly intersecting lines in the shape of a cross. Other variations of the reticle include dots, posts, chevrons, circles, squares, thruCross, barCross, etc.

[0040] At block **315**, GPU **225** may output a content to be displayed, such as the video frame, and transmit the video frame to scalar **265**. At block **320**, scalar **265** may receive the content from GPU **225**. The method may proceed to decision block **325** where scalar **265** may determine whether the adaptive reticle function over the video is enabled. If the reticle function over the video frame is enabled, then the “YES” branch is taken, and the method proceeds to block **330**. If the reticle function over the video frame is disabled or deactivated by the user, then the “NO” branch is taken, and the method ends.

[0041] At block **330**, scalar **265** or OSD engine **275** in particular may render the reticle over the content based on the settings selected by the user. The reticle may be rendered initially based on the user-selected settings. The method may proceed to block **335** where scalar **265** or DCR engine **270** in particular, may obtain samples of RGB values from pixels in a sample area. The sample area can be a square or any other form of pixel block underneath the reticle. Scalar **265** or DCR engine **270** may then calculate an average of the RGB values of the pixels in real-time. The size of the sample may be based on the selected reticle. For example, for a default crosshair with a bar-cross shape, the sample collected may have a size of 50×50 pixels. Accordingly, scalar **265** or DCR engine **270** in particular may output an average of the RGB values of 2500 pixels.

[0042] The method may proceed to block **340** where scalar **265** or MCU engine **280** in particular may compare the average of the RGB values with the RGB value of the reticle. The method may proceed to decision block **345** where scalar **265** or DCR engine **270** in particular may determine whether a color difference, also referred to as delta empfindung (delta E), between the color of the reticle and the average of the RGB values is less than a threshold. A delta E between three and six is generally considered an acceptable number in commercial reproduction. In particular, most people can distinguish the difference when the delta E is above five. As such, the threshold may be set to a value between three and six, such as five. However, the threshold may be set to a different number. The threshold can also be set to different individual values for R, G, and B. If the color difference is less than the threshold, then the “YES” branch is taken, and the method proceeds to

block **350**. If the color difference is not less than the threshold, then the “NO” branch is taken, and the method proceeds to block **355**.

[0043] At block **350**, scalar **265**, or OSD engine **275** in particular, may render the reticle with a color that is complementary to the average RGB values over the video frame. The complementary color may be chosen to create a contrast to the average RGB values for a striking optical effect, which allows the reticle to be “always visible.” At block **355**, scalar **265**, or OSD engine **275** in particular, may render the reticle with the user-selected color over the video frame.

[0044] FIG. **4** shows a display device **400** with a reticle **410** superimposed over an image **405**. In this example, the color of reticle **410** may change based on an average color of image **405** underneath. In particular, the color of reticle **410** may be a complementary color of the average color. In addition, the color of reticle **410** may change adaptively as reticle **1010** is moved around, and the average color of the image underneath changes. This color inversion of the reticle is also depicted in FIG. **5** which shows a display device **500** with a reticle **510** superimposed over an image **505**.

[0045] FIG. **6** shows a system **600** for an adaptive display of an on-screen reticle. System **600** includes an information handling system **605** that is coupled to a display device **640** via an interface **690**. Information handling system **605**, which is similar to information handling system **205** of FIG. **2**, includes a memory **610**, a CPU **620**, and a GPU **625**. Memory **610** includes an operating system **615**. Display device **640** includes a display panel **645** and an interface board **660**. Display panel **645** includes a timing control unit **650**. Interface board **660** includes a scalar **665**. Scalar **665** includes a primary video path **675** and a secondary video path **680**. Display panel **645** is similar to display panel **245** of FIG. **2**. Timing control unit **650** is similar to timing control unit **250** of FIG. **2**. Scalar **665** is similar to scalar **265** of FIG. **2**. Memory **610** is similar to memory **210** of FIG. **2**. CPU **620** is similar to CPU **220** of FIG. **2**. GPU **625** is similar to GPU **225** of FIG. **2**. The components of system **600** may be implemented in hardware, software, firmware, or any combination thereof. The components shown are not drawn to scale and system **600** may include additional or fewer components. In addition, connections between components may be omitted for descriptive clarity.

[0046] Display device **640** is similar to display device **240** of FIG. **2**. However, display device **640** also has a picture-in-picture (PIP) and/or picture-by-picture (PBP) capability which allows a primary content to be presented via a primary window while secondary content may be presented via a secondary window. In one example, the secondary window may be the PIP or PBP window that can be superimposed or overlaid on the primary window. A primary video path **675**, also referred to as a main video path may be used for displaying a primary content, such as content from GPU **625**. The content from GPU **625** can be an image, video frame, video stream, or similar. A secondary video path also referred to as an auxiliary video path, may be used for displaying auxiliary video content, also referred to as the secondary video content. Accordingly, scalar **665** may be configured to leverage the PIP/PBP functionality of display device **640** and present a reticle in the secondary window via the secondary video path **680**. For example, the reticle may be rendered as an overlay PIP window on the primary content. The primary content may be presented via primary video path **675**. Thus, scalar **665** may be configured to leverage the utilization of the dual video path process function when the visible display on-screen reticle is enabled.

[0047] In one embodiment, the reticle may be presented using a filtered mask layer as an overlay on the primary content creating a visual effect. In particular, scalar **665** may be configured to create a mask layer that adopts the reticle's contour. The mask layer can be combined with a filter that includes the reticle. The combination may be rendered at the secondary window via secondary video path **680** and superimposed over the primary content or a portion thereof which is rendered at the primary window via primary video path **675**. For example, the combination may be rendered using the PIP/PBP functionality of display device **640**. The filtered mask layer may provide an exclusive OR (XOR) function resulting in color inversion. In particular, the combination results in

a pixel-by-pixel visual effect, wherein RGB values of the pixels of the reticle may change to complement the RGB values of the pixels underneath the reticle, such as depicted in images **725** and **735**.

[0048] FIG. **7** shows a composite **700** that includes a series of images for an adaptive display of an on-screen reticle. Composite **700** includes an image **705**, a mask **710**, a masked image **715**, a filter **720**, an image **725**, and an image **735**. Image **705** may be an image presented via a primary video path. Mask **710** and filter **720** may be linked or combined, such that when the on-screen reticle is moved, mask **710** and filter **720** may move together. Image **705** may be primary content, such as a scene generated by an application like a gaming application. Image **705** includes a background **745** and an object **740**, wherein the color of background **745** may be different than the color of object **740**. In this example, object **740** has an RGB color green (0×00FF00).

[0049] Mask **710** may be a mask layer of an area based on the size of the reticle, wherein the area of the mask layer is greater than the area of the reticle. Further, image **705** may have an area that is larger than mask **710**. Mask **710** includes a pattern of a pre-defined shape of the reticle, as selected by the user. In this example, mask **710** includes a pre-defined crosshair with a bar-cross shape. The pattern may be translucent on an opaque base. Accordingly, mask **710** may be applied to image **705**, such that the color of pixels underneath mask **710** may be concealed except for the pixels underneath the pattern. For example, mask **710** may be overlaid on image **705** resulting in masked image **715**. In this example, masked image **715** shows a portion of background **745** and a portion **750** of object **740** through the pattern. As such, portion **750** may include pixels of object **740** underneath the reticle pattern of mask **710**.

[0050] Filter **720**, with an area that is similar to the area of mask **710**, may be applied to masked image **715** resulting in image **725**. For example, filter **720** may be overlaid or superimposed on masked image **715**. Filter **720** may include an RGB white (0×FFFFFF) as a base layer **755** and a reticle **760** which is based on the user-selected shape and color. The color and shape of the reticle may be preselected by the user during the activation of the reticle function over the video frame. In this example, reticle **760** has a color similar to the RGB color green (0×00FF00). Because object **740** and reticle **760** have similar colors, a portion of reticle **760** that overlaps portion **750** may “disappear” against the backdrop. To mitigate this issue, filter **720** may have been designed to change the colors of the pixels of the portion of reticle **760** that overlaps portion **750** to complement the colors of the pixels underneath said portion of reticle **760** resulting in a desired multi-color effect. This may increase the visibility of the overlapping portion of reticle **760** with portion **750**. For example, the combination of RGB white (0×FFFFFF) of filter **720** when overlaid against the RGB green (0×00FF00) of portion **750** of object **740** may result in an XORed value of RGB magenta (0×FF00FF) of portion **770** of reticle **760**, which is a complementary color of RGB green (0×00FF00). Portion **770** is a portion of reticle **760** that may be superimposed over portion **750**. In this example, the color of portion **770** may be complementary to the color of portion **850**. Portion **770** is shown with the visual effects. This visual effect of portion **770** of reticle **860** over image **705** is also depicted in image **735**. This visual effect of reticle **760** over image **705** is depicted in image **735**.

[0051] FIG. **8** shows a composite **800** that includes a series of images for an adaptive visible display on-screen reticle. Composite **800** includes an image **805**, a mask **810**, a masked image **815**, a filter **820**, an image **825**, and an image **835**. Similar to above, mask **810** and filter **820** may be linked, such that when the on-screen reticle is moved, mask **810** and filter **820** may move together. Image **805** may be an image, scene, video stream, etc. that is generated by an application, such as a gaming application, and rendered via a primary video path. Image **805** includes a background **830**, an object **840**, and an object **845**, wherein the color of background **830** may be different than the color of objects **840** and **845**. In this example, object **845** has an RGB color green (0×00FF00) while object **840** has an RGB color black (0×000000).

[0052] Mask **810** may be a mask layer with an area generally larger than the size and/or area of the

reticle. Further, image **805** may have an area that is generally larger than the area of mask **810**, wherein the area of image **805** may be up to the resolution of the display. Mask **810** may include a base and a pattern of a pre-defined shape of the reticle, as selected by the user. In this example, the pattern is a crosshair with a bar-cross shape. Similar to mask **810**, the base of mask **810** may have an opaque color such that the color of the pixels underneath the portion of mask **810** with the opaque color may be concealed. In addition, similar to mask **810**, the pattern may have a transparent color such that the color of the pixels underneath the pattern may be revealed. In this example, mask **810** may be overlaid on image **805** resulting in masked image **815**. Masked image **815** may reveal a portion **855** of object **845**, a portion **850** of object **840**, and a portion of background **830** underneath the pattern. As such, portion **850** may reveal the colors of the pixels of object **840** while portion **855** may reveal the colors of the pixels of object **845**.

[0053] Filter **820**, with an area that is similar to the area of mask **810**, may be applied to masked image **815** resulting in image **825**. For example, filter **820** in combination with mask **810** may be overlaid on masked image **815**. Filter **820** may include a base **865** with an RGB white ($0 \times \text{FFFFFF}$) and a reticle **860** according to the user-selected shape and color. The color and shape of the reticle may be selected by the user during the activation of the reticle function over the video frame. In this example, reticle **860** has a color similar to the RGB color black (0×000000) of object **840**. Because object **840** and reticle **860** have similar colors, a portion of reticle **860** that overlaps portion **850** may “disappear” against the backdrop.

[0054] To mitigate this issue, filter **820** may have been designed to complement the colors of the pixels underneath reticle **860** resulting in a desired multi-color effect. This may then increase the visibility of the overlapping portion of reticle **860** with portion **850**. For example, the combination of RGB white ($0 \times \text{FFFFFF}$) of filter **820** when overlaid against the RGB black (0×000000) of object **840** may result in an XORed value of RGB white ($0 \times \text{FFFFFF}$) for portion **870**, which is a complementary color of RGB black (0×000000). Portion **870** is a portion of reticle **860** that may be superimposed over portion **850**. In this example, the color of portion **870** may be complementary to the color of portion **850**. Portion **870** is shown with the visual effects. This visual effect of portion **870** of reticle **860** over image **805** is also depicted in image **835**. Because the color of portion **855** is different than object **845**, the color of pixels associated with portion **855** may not be inverted as depicted in portion **875**.

[0055] FIG. **9** shows a flowchart of a method **900** for an adaptive display of an on-screen crosshair. Method **900** may be performed by any suitable component of system **600** of FIG. **6**, including, but not limited to scalar **665**. While embodiments of the present disclosure are described in terms of the components of system **600** of FIG. **6**, it should be recognized that other components may be utilized to perform the described method. Method **900** may be performed while an adaptive reticle function over a video frame is enabled by a user.

[0056] Method **900** typically starts at block **905** where scalar **665** creates a mask layer with a predefined reticle, which may mask out images in the background outside of the reticle. The shape and/or color of the reticle may be selected by the user. For example, the user may use the provided default shape and color. In another example, the user may select a shape and color other than the provided defaults. The method may proceed to block **910**.

[0057] At block **910**, scalar **665** may create a filter layer with a pre-defined reticle. The filter layer may be combined with the mask layer to perform an XOR logic function at block **915**. The method may proceed to block **920** where scalar **665** may overlay the mask with the filter layer to an image base layer. The method may then proceed to block **925** where scalar **665** may present a visual effect of the superimposed combination mask and filter over the image base layer in real-time.

[0058] FIG. **10** shows a display device **1000** with a reticle **1010** superimposed over an image **1005**. In this example, the color of reticle **1010** may change pixel by pixel based on the color of the primary content underneath, wherein the color of the pixels of reticle **1010** may be complementary to the color of the image's pixels underneath. For example, in areas where the color of the image is

white, the color of a portion of the reticle over the said area is black. Accordingly, in areas where the color of the image is black, the color of the portion of the reticle over the said area is white. In addition, the color of the pixels of reticle **1010** may change adaptively as reticle **1010** is moved around. This pixel-by-pixel color inversion of the reticle is also depicted in FIG. **10** which shows a display device **1100** with a reticle **1110** superimposed over an image **1105**.

[0059] Although FIG. **3**, and FIG. **9** show example blocks of method **300** and method **900** in some implementations, method **300** and method **900** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **3** and FIG. **9**. Those skilled in the art will understand that the principles presented herein may be implemented in any suitably arranged processing system. Additionally, or alternatively, two or more of the blocks of method **300** and method **900** may be performed in parallel. For example, blocks **330** and **335** of method **300** may be performed in parallel.

[0060] In accordance with various embodiments of the present disclosure, the methods described herein may be implemented by software programs executable by a computer system. Further, in an exemplary, non-limited embodiment, implementations can include distributed processing, component/object distributed processing, and parallel processing. Alternatively, virtual computer system processing can be constructed to implement one or more of the methods or functionalities as described herein.

[0061] When referred to as a “device,” a “module,” a “unit,” a “controller,” or the like, the embodiments described herein can be configured as hardware. For example, a portion of an information handling system device may be hardware such as, for example, an integrated circuit (such as an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a structured ASIC, or a device embedded on a larger chip), a card (such as a Peripheral Component Interface (PCI) card, a PCI-express card, a Personal Computer Memory Card International Association (PCMCIA) card, or other such expansion card), or a system (such as a motherboard, a system-on-a-chip (SoC), or a stand-alone device).

[0062] The present disclosure contemplates a computer-readable medium that includes instructions or receives and executes instructions responsive to a propagated signal; so that a device connected to a network can communicate voice, video, or data over the network. Further, the instructions may be transmitted or received over the network via the network interface device.

[0063] While the computer-readable medium is shown to be a single medium, the term “computer-readable medium” includes a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term “computer-readable medium” shall also include any medium that is capable of storing, encoding or carrying a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

[0064] In a particular non-limiting, exemplary embodiment, the computer-readable medium can include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable medium can be a random-access memory or other volatile re-writable memory. Additionally, the computer-readable medium can include a magneto-optical or optical medium, such as a disk or tapes, or another storage device to store information received via carrier wave signals such as a signal communicated over a transmission medium. A digital file attachment to an e-mail or other self-contained information archive or set of archives may be considered a distribution medium that is equivalent to a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium or a distribution medium and other equivalents and successor media, in which data or instructions may be stored.

[0065] Although only a few exemplary embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the

embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents but also equivalent structures.

Claims

1. A method comprising: obtaining, by a processor, red, green, and blue (RGB) values of pixels from an area underneath a reticle; calculating an average of the RGB values of the pixels from the area underneath the reticle; and if a difference between the average of the RGB values and an RGB value of the reticle is less than a threshold, then changing the RGB value of the reticle to a complementary color for display.
2. The method of claim 1, further comprising comparing the average of the RGB values with the RGB value of the reticle.
3. The method of claim 1, further comprising if the difference between the average of the RGB values and the RGB value of the reticle is less than the threshold, then displaying the reticle using the RGB value.
4. The method of claim 1, wherein the RGB value of the reticle is selected by a user.
5. The method of claim 1, wherein the area includes 50 pixels by 50 pixels.
6. The method of claim 1, wherein the threshold is a delta empfindung value.
7. The method of claim 1, wherein the method is performed subsequent to a user activating the reticle.
8. A display device, comprising: a display panel configured to display a reticle; and a scalar configured to: obtain red, green, and blue (RGB) values of pixels from an area underneath the reticle; calculate an average of the RGB values of the pixels from the area underneath the reticle; and if a difference between the average of the RGB values and an RGB value of the reticle is less than a threshold, then display the reticle using a complementary color of the RGB value.
9. The display device of claim 8, wherein the scalar is further configured to compare the average of the RGB values with the RGB value of the reticle.
10. The display device of claim 8, wherein the scalar is further configured to determine if the difference between the average of the RGB values and the RGB value of the reticle is less than the threshold, then displaying the reticle using the RGB value.
11. The display device of claim 8, wherein the RGB value of the reticle is selected by a user.
12. The display device of claim 8, wherein the threshold is a delta empfindung value of five.
13. A method comprising: receiving, by a processor, an image for display; superimposing a mask over the image resulting in a masked image; and applying a filter that includes a reticle of a red, green, and blue (RGB) value to the masked image resulting in inversion of RGB values of pixels of the reticle that overlaps the image.
14. The method of claim 13, wherein the mask includes pre-defined contours of the reticle.
15. The method of claim 13, wherein the inversion of the RGB values of the pixels of the reticle is performed pixel by pixel.
16. The method of claim 13, wherein the inversion of the RGB values of the pixels of the reticle that overlaps the image resulting in a visual effect unto the reticle.
17. The method of claim 13, wherein the inversion of the RGB values of the pixels is performed in real-time.
18. The method of claim 13, wherein a base of the filter has an RGB value of white.
19. The method of claim 13, wherein the mask and the filter are displayed at a picture-in-picture

window.

20. The method of claim 13, wherein the image is displayed via a main video path.
